



Extreme Heat, Health Equity, and Student Well-being

Abstract

Hotter temperatures are more than uncomfortable: they can make children seriously ill. This lesson, based on a Harvard Chan C-CHANGE guide, explains what heat-related illness is, why climate change is bringing more dangerous hot days, which children face the greatest risk, and how families, schools, and coaches can keep young people safe.

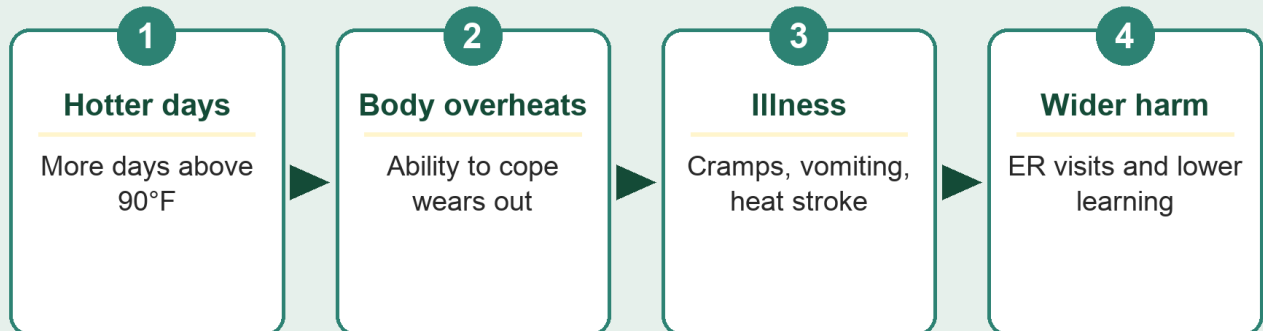
Introduction

Heat-related illness happens when the body severely overheats, a bit like having a fever without an infection. When a child is exposed to high temperatures for a long time, the body's ability to cope wears out. Symptoms range from headaches and muscle cramps to vomiting, confusion, and in extreme cases unconsciousness. The most serious form, heat stroke, occurs when body temperature rises to 104°F (40°C) and can damage the brain, heart, kidneys, and muscles. Heat waves have always been part of summer in much of the United States, but climate change is making them hotter, more frequent, and more intense, putting more children at risk.

How Do Scientists Study This?

Researchers study heat and children's health by linking temperature data to health records. For example, studies compare emergency department visits on hot summer days with cooler days to see which illnesses increase. National databases track deaths from vehicular heat stroke and heat illnesses among high school athletes. Climate projections estimate how many more days above 90°F different regions will experience. Researchers also analyze how heat exposure differs across neighborhoods and populations, including urban heat islands shaped by past policies. Government assessments, such as a 2023 EPA report, combine this evidence to estimate effects on learning and health.

How extreme heat harms children



Link heat to health

Compare emergency visits on hot and cooler days to see which illnesses rise.

Identify who is at risk

Track groups most affected, such as young children, athletes, and kids in heat islands.

Project future heat

Estimate how many more hot days regions will see as carbon pollution builds.

What Did They Discover?

On hot summer days, children are more likely to visit emergency departments for heat illness and also for intestinal infections, ear infections, nervous system diseases, and other reasons. If carbon pollution keeps building, most of the U.S. could see 20 to 30 more days above 90°F each year, and the Southeast 40 to 50. Over 20 years, an average of 37 children died each year of vehicular heat stroke, 87% of them younger than 3. About 9,000 high school athletes are treated for exertional heat illness each year, with football players at highest risk. Heat also harms learning, with a projected 4% to 7% yearly drop in achievement, and falls hardest on children in urban heat islands.

Heat and kids by the numbers

20–30 more days above 90°F each year for most of the U.S.

37 children a year died of vehicular heat stroke

9,000 high school athletes treated for heat illness yearly

Some common medications, including diuretics, diphenhydramine, and many mental health medicines, can impair the body's ability to cope with heat.

Takeaways

Heat is dangerous

Heat stroke can damage vital organs and requires emergency treatment.

Risk is unequal

Young children, athletes, and kids in heat islands face higher risks.

Prevention works

Water, light clothing, cool spaces, and never leaving kids in cars save lives.

Key Terms

Heat-related illness

Sickness caused when the body overheats, ranging from cramps and exhaustion to life-threatening heat stroke.

Heat stroke

The most serious heat illness, when body temperature reaches 104°F (40°C) or higher and organs can be damaged.

Exertional heat illness

Heat illness brought on by intense physical activity, such as sports practice in hot weather.

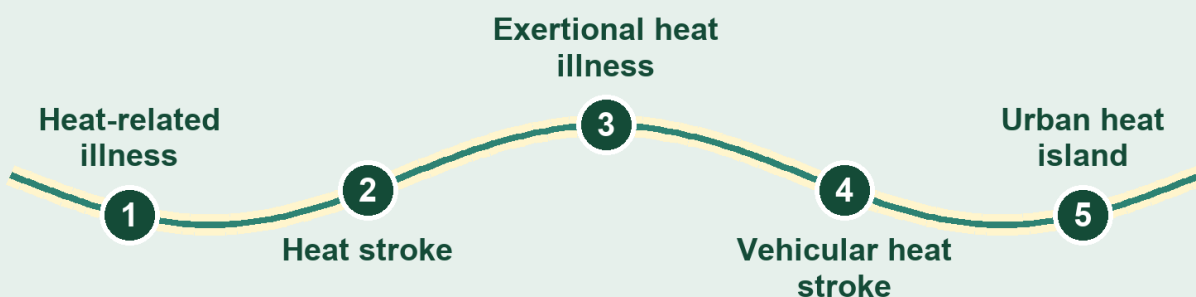
Vehicular heat stroke

Heat stroke that occurs when a child is left in a hot vehicle, where temperatures rise quickly.

Urban heat island

A city area that is hotter than surrounding areas because of pavement, buildings, traffic, and little green space.

Vocabulary Path · Extreme Heat, Health Equity, and Student Well-being



Activity

Heat Safety Plan for Athletes

Students design a heat safety protocol for a school sports team and a poster to share it.

Materials: Protocol template; Local weather forecast; Poster paper and markers

1. List the warning signs of heat-related illness from the lesson.
2. Decide on rules for practice on hot days, such as timing, water breaks, and rest in the shade.
3. Write emergency steps to follow if an athlete shows signs of heat stroke.
4. Explain which athletes might be at higher risk and how the plan protects them.
5. Create a poster that summarizes your protocol for coaches and players.

References

Harvard T.H. Chan School of Public Health, C-CHANGE. Climate, Kids, and Health: Heat and Health.
<https://hsph.harvard.edu/research/climate-health-c-change/climate-kids-and-health/heat-and-health/>